R7.7 – the trace pair as a coset label for a = 1 and for E/8E

Curves $y^2 + xy = x^3 + ax^2 + b$ over $\mathbb{F}_{2^n}$, n odd, $b \neq 0$. Toy fields ($n \leq 23$); no claim about any deployed curve. Run records: `exhaustive.json`, `exhaustive_alla.json`, `random.json`, `control*.json`, `cost.json`, `witness.json`, `search_level*.json`; experiment `examples/r7_7_coset_label.rs`; full text in `RESULTS.md`.

Verdicts

Hypothesis	Statement and outcome	Verdict
H1 (as posed)	For $\mathrm{Tr}(a) = 1$, $E/4E$ has R6.3′s structure and the trace pair labels it. False premise: $v_2(\#E) = 1$, $2E$ has odd order, $E/4E = E/8E = E/2E = \mathbb{Z}/2$. The label is $\mathrm{Tr}(x) + \mathrm{Tr}(a)$ (0 wrong in 148,598,015 points). The second bit satisfies $b_2(-P) = b_2(P) + \mathrm{Tr}(x)$, so on $2E$ it flips between P and $-P$ in one coset: the pair is non-constant on 312,667 of 312,684 cosets (the 17 exceptions are tiny $n = 5$ cosets). 0 cross-coset collisions.	**refuted as posed; corrected form confirmed**
H1′ (R6.3 for any a, b with $\mathrm{Tr}(a) = 0$)	$(\mathrm{Tr}(x), b_2)$ is constant on every coset of $4E$ and equals $(0,0), (0,1), (1,\beta), (1,1+\beta)$ for $k = 0, 2, 1, 3$, with $\beta = \mathrm{Tr}(bH(b))$ ($= \frac{n+1}{2} \bmod 2$ at $b = 1$). 0 failures in 71.6 M, 68.0 M and 9.0 M points (exhaustive $n \leq 13$; all (a, b), $n \leq 9$; random $n = 17..23$).	**confirmed** (constancy measured, rest proved)
H2 ($E/8E$)	$\mathbb{Z}/4 \times \mathbb{Z}/2$ is impossible ($E(F)[2] = \{O, T\}$, 2-part cyclic). The pair cannot label $E/8E$ (4 values, 8 cosets; smallest counterexample $n = 5$, $a = 0$, $b = 6$). With $b_3(R) = b_2(R/2)$ for even R and $b_3(P) = b_3(P + Q_0)$ for odd P, the triple is a bijection $E/8E \to \mathbb{F}_2^3$; the same recursion labels $E/16E$, $E/32E$. 0 failures in 74.1 M / 37.6 M / 18.2 M points. No closed trace formula for odd points was found among 120 trace monomials.	**confirmed; datum not cheap**
CT bar ($< 2\times$)	$\mathrm{Tr}(a) = 1$: one masked trace, ratio 1 by construction. $\mathrm{Tr}(a) = 0$: no new half-trace algorithm, not re-timed; R6.3′s 2.5–5.0 × wall, 2.0–3.7 × words stands.	**met (a=1); not met (a=0)**

The label tables

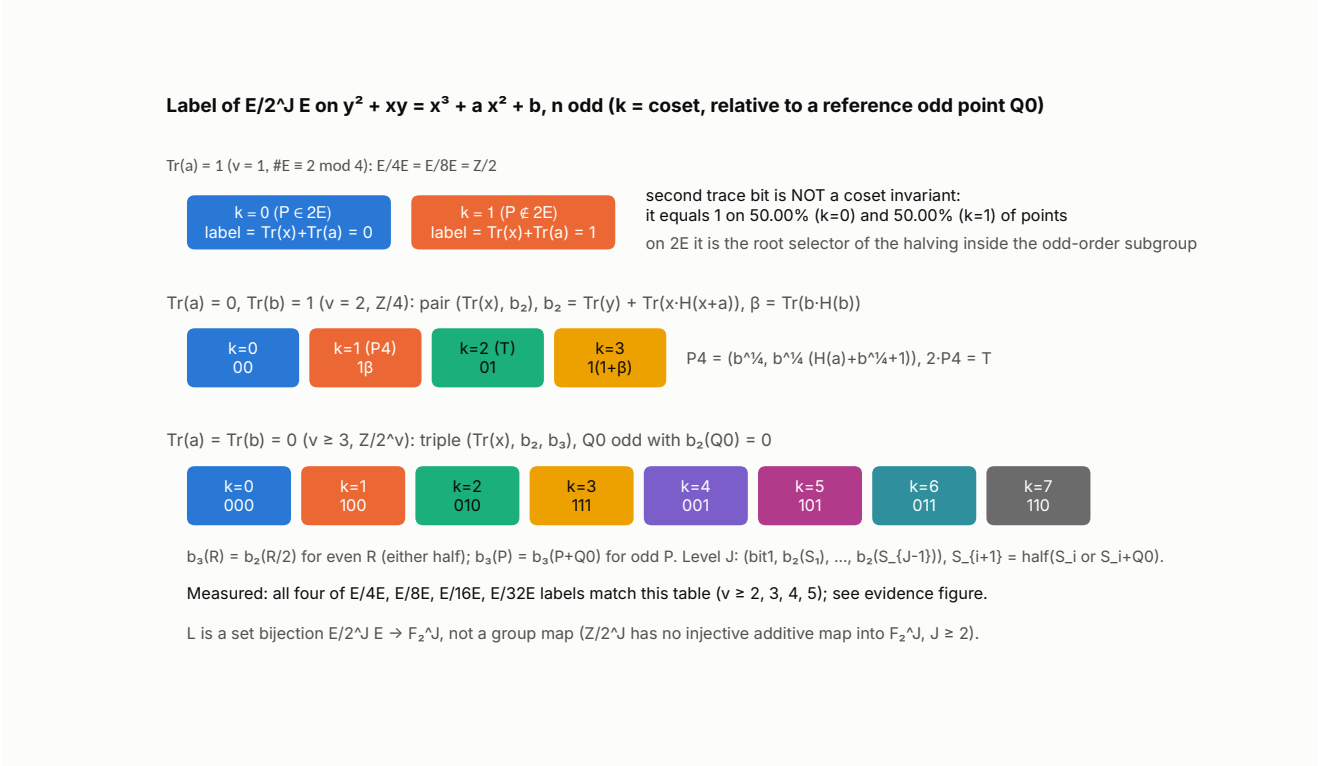


Figure 1: Label of $E/2^J E$ by case; k counts from a reference odd point Q_0 ($b_2(Q_0) = 0$ for $v \geq 3$; P_4 for $v = 2$). `plot_data.json`.

Evidence

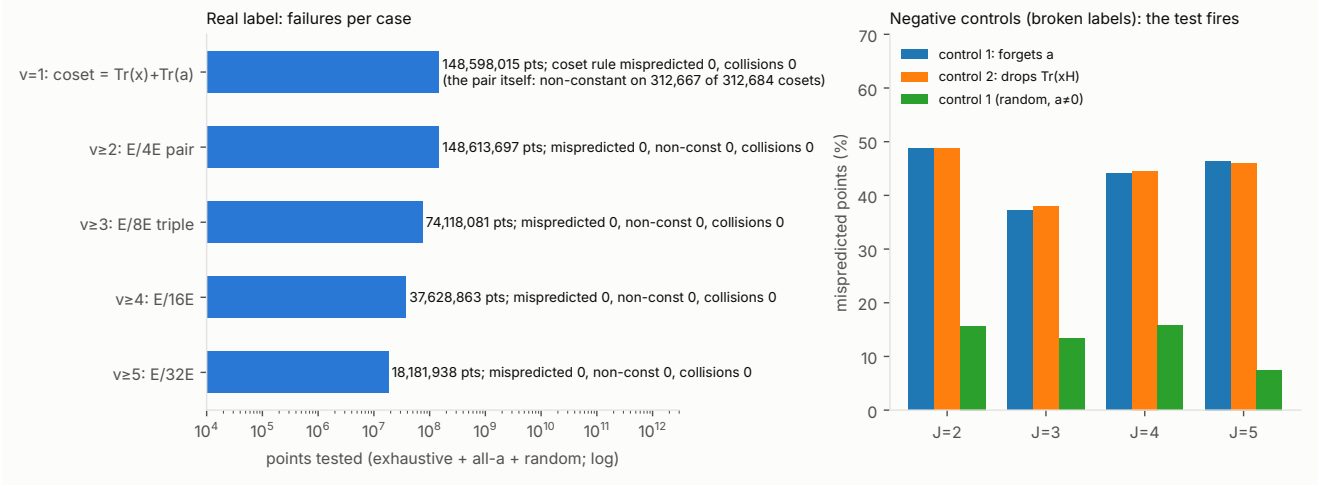


Figure 2: Left: points tested per case (exhaustive $n \leq 13$, all-a $n \leq 9$, random $n = 17..23$ summed, overlap at $n \leq 9$) with failures; right: two deliberately broken labels fail in the same code (negative controls).

Proved: structure ($P \in 2E \Leftrightarrow \mathrm{Tr}(x) = \mathrm{Tr}(a)$, v from $\mathrm{Tr}(a), \mathrm{Tr}(b)$), the $a = 1$ statements, the $4E$ criterion for all a, b, $\beta = \mathrm{Tr}(bH(b))$, x-only forms $\mathrm{Tr}(b/x^2) = \mathrm{bit}_1$ and $b_2 = \mathrm{Tr}(xH(b/x^2))$ for even P. Measured only: constancy of b_2 on odd cosets for general (a, b) (R6.3 proves $(0, 1)$) and the $J \geq 3$ tables that rest on it.

Cost

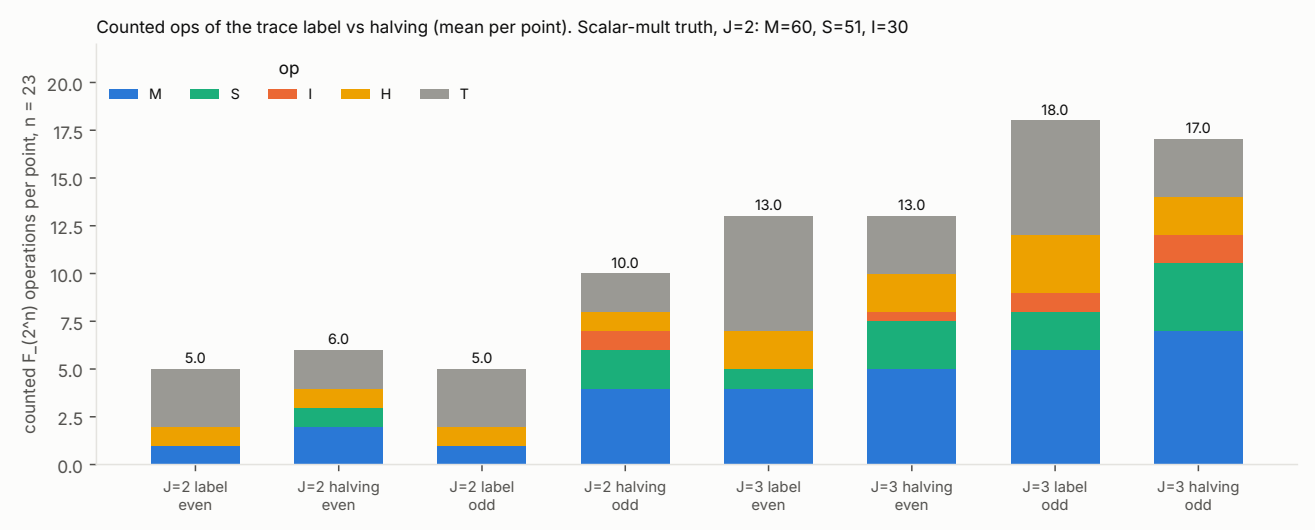


Figure 3: Counted $\mathbb{F}_{2^n}$ operations per point, $n = 23$ (M product, S squaring/sqrt, I inversion, H half-trace, T trace); `cost.json`, 24,000 points per bar. Scalar-multiplication truth for comparison is about 60 M + 50 S + 30 I.

The pair ($J = 2$) saves the point addition of the halving route on odd points ($I + 3M + 2S$) and $M + S$ on even points for one more trace. For $J \geq 3$ the label is not cheaper than halving. Wall time was not measured.

Limits

Toy fields, n odd only; exhaustive to $n = 13$ ($a \in \{0, 1\}$) and $n = 9$ (all a), sampled to $n = 23$ (1500 points x 1000 curves per n and a-class). The 120-monomial search is a finite family. Counted operations, not weights. Constancy for general (a, b) is verified, not proved.